

Material Datasheet

Material Type

Metal

Material Sub-Type

Stainless Steel

Material Name

Stainless Steel 15-5 PH



Note: Properties are not guaranteed. They are intended only as a basis for comparison and not for design purposes.

Mechanical Properties

Yield Tensile Strength	976 MPa	141.5 Ksi
Ultimate Tensile Strength	1114 MPa	161.5 Ksi
Tensile Modulus	200 GPa	29.0 Msi
Shear Strength	896 MPa	130.0 Ksi
Shear Modulus	76 GPa	11.0 Msi
Fatigue Strength	655 MPa	95.0 Ksi
Elongation At Break	8 %	-
Poisson's Ratio	0.28	-
Hardness	C36/363	-
Density	7.8 g/cm ³	0.282 lb/in ³

Electrical/Magnetic Properties

Surface Resistance (ANSI)	77.0 μΩ	-
Magnetism	High	-

Thermal Properties

Thermal Expansion Coefficient	10.8 μm/(m·°C)	6.0 μin/(in·°F)
Specific Heat Capacity	0.420 J/g·°C	0.100 BTU/lb·°F
Thermal Conductivity	17.8 W/(m·°C)	10.3 BTU/(h·ft·°F)
Melting Point	1405 °C	2561 °F

Chemical Properties

Corrosion Resistance	Excellent	-
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